

Final Review: Every Trap and Mixed Exam Practice

Weeks 1–12 in one deck: trap sheets, which method when, what to memorize, and 12 exam-style problems

Use this the week before the midterm (weeks 1–6) and the final (everything).
Do the 12 problems under a timer: about 12 minutes each.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed.

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

By **Vladislav Shkuratov**, made with Claude (AI). Independent study notes, not affiliated with Concordia University or any instructor.

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Trap sheet: week 1 (circuit concepts)

Top traps

- One wire = one node, however it's drawn. Color the nodes before combining anything.
- Series needs **nothing else** attached to the shared node.
- Shorted resistor: gone. Dangling resistor: carries nothing. Closed switch: shorts everything in parallel with it.
- Current divider uses the **other** resistor (two branches only).
- A divider only works if nothing else is attached to the middle node (the loaded divider).
- Sources can absorb power. Use $p = vi$ with i into $+$ and trust the sign.
- R across a V source and R in series with an I source don't affect the rest of the circuit.

Trap sheet: week 2 (nodal and mesh)

Top traps

- Use the method the question asks for, or risk 0 marks.
- V source to ground: known node, no KCL there. Floating V source: supernode (one KCL + the source equation).
- I source on an outer edge: known mesh. Shared I source: supermesh (source equation + outer KVL).
- Shared resistor in mesh: own current minus the neighbour's.
- Dependent source: write its control variable in your unknowns. The matrix is no longer symmetric.
- Write equations with variables first, then plug numbers in.

Trap sheet: week 3 (Thévenin, Norton, max power)

Top traps

- Remove the load before finding V_{oc} and R_{Th} .
- Kill only **independent** sources: $V \rightarrow$ wire, $I \rightarrow$ gap. Never kill a dependent source.
- With a dependent source: test source, or $R_{Th} = V_{oc}/I_{sc}$.
- Superposition adds voltages and currents, never powers.
- Source transformation: the current arrow points toward the V source's $+$.
- $P_{max} = V_{Th}^2/4R_{Th}$, at 50% efficiency. Draw the equivalent when asked.

Trap sheet: week 4 (L, C, transients)

Top traps

- DC steady state: C is a gap, L is a wire. “Energy stored” at DC needs no differential equations.
- Capacitors combine the opposite way to resistors.
- Only v_C and i_L are continuous. i_C and v_L can jump (the inductor kick).
- $\tau = R_{Th}C$ or L/R_{Th} , using the circuit **after** the switch, seen from the C or L.
- First order: $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$. Show $x(0^+)$, $x(\infty)$, τ separately.
- Second order: compare α with ω_0 . Needs two initial conditions.

Trap sheet: week 5 (sinusoids)

Top traps

- Convert to positive cosines before comparing phases:
 $\sin x = \cos(x - 90^\circ)$, $-\cos x = \cos(x \pm 180^\circ)$.
- ω (rad/s) vs f (Hz). $377 \leftrightarrow 60$ Hz.
- RMS of a sinusoid is $V_m/\sqrt{2}$, but **not** for other waveforms.
- $\tan^{-1}$ quadrant: $a < 0$ means add 180° . Sketch the point.
- Degrees vs radians in the calculator.

Trap sheet: week 6 (phasors, impedance)

Top traps

- $\mathbf{Z}_C = -j/(\omega C)$: don't lose the minus sign.
- Magnitudes don't add. $|\mathbf{V}_L|$ can exceed the source voltage.
- Inductive or capacitive: the sign of X **after** combining.
- $G \neq 1/R$ for a complex impedance.
- Different frequencies: separate solutions, add in the time domain.
- Give $v(t)$ with ω when asked for the time domain.

Trap sheet: week 7 (AC power)

Top traps

- Peak values need $\frac{1}{2}$ in P . RMS values don't.
- $\mathbf{S} = \mathbf{VI}^*$: conjugate the current.
- $\theta = \theta_v - \theta_i$ = the load impedance angle.
- Say lagging or leading. Units: W, var, VA.
- Add P and Q separately, never $|\mathbf{S}|$.
- Correction capacitor: parallel, V_{rms} , ω in rad/s; P unchanged.
- Max power: $\mathbf{Z}_L = \mathbf{Z}_{Th}^*$; resistive only: $R_L = |\mathbf{Z}_{Th}|$.

Trap sheet: week 8 (transformers)

Top traps

- Check which ratio “ n ” or “ N ” means in the question.
- Voltage scales with the ratio, current with its inverse, impedance with its **square**.
- Dots only change sign, never magnitude.
- No DC through a transformer.
- Matching: C cancels the reactance, N matches the resistance. Convert peak sources to RMS.

Trap sheet: week 9 (three-phase)

Top traps

- A “480 V system” means the RMS **line** voltage.
- Y: $V_L = \sqrt{3}V_p$, $I_L = I_p$. Δ : $V_L = V_p$, $I_L = \sqrt{3}I_p$.
- The pf angle is the per-phase impedance angle.
- $P = \sqrt{3}V_L I_L \cos \theta$ is the total. Don't write $3V_L I_L$.
- Per-phase method: convert Δ with $\mathbf{Z}_{\Delta}/3$ first.

Trap sheet: week 10 (magnetic circuits)

Top traps

- Convert cm and cm^2 to m and m^2 first.
- Gap: $\mu_r = 1$. The gap usually dominates the reluctance and stores most of the energy.
- MMF is NI . $L = N^2/\mathcal{R}$.
- Nonlinear core: read the B–H table; doubling I doesn't double the flux.
- Boundaries: normal B and tangential H are continuous.

Trap sheet: weeks 11–12 (energy conversion, machines)

Top traps

- ω in rad/s: $\omega = 2\pi n/60$. 1 hp = 746 W.
- Rated power is **output**: $\text{input} = P_{\text{rated}}/\eta$.
- Shunt motor: $I_L = I_f + I_a$; only I_a goes through R_a .
- Motor $V = E_a + I_a R_a$; generator $V = E_a - I_a R_a$.
- Constant flux: $E_a \propto \text{speed}$. Use ratios.
- $n_s = 120f/p$ with p = number of poles. Slip is a fraction of n_s .
- Starting: $E_a = 0$, current limited only by R_a .

Which method when: DC circuits

If the question...	use...
asks for R_{eq}	color nodes, collapse from the far end, redraw each step
is a ladder	collapse, or “assume $I_x = 1$ ” and scale
names a method	that method, no matter what
has many current sources	nodal
has many voltage sources	mesh
asks about one load	Thévenin (then max power if asked)
has several independent sources	superposition, or just nodal/mesh
has a balanced bridge	delete the middle resistor

Which method when: transients and AC

If the question...	use...
has L/C and a DC source “for a long time”	DC steady state (C gap, L wire)
has a switch at $t = 0$ with one L or C	the first-order recipe
has one L and one C	α vs ω_0 , two initial conditions
has sinusoidal sources	phasors + impedances + week 1–3 methods
mixes frequencies or DC + AC	superposition, add in time
asks for P , Q , pf	$\mathbf{S} = \mathbf{VI}^*$, or $I^2 R$ and $I^2 X$
gives loads as “kW at pf”	$Q = P \tan(\cos^{-1} \text{pf})$; add P 's and Q 's

Which method when: transformers, 3-phase, magnetics, machines

If the question...	use...
has a transformer	reflect the load: $\mathbf{Z}_L(N_1/N_2)^2$
asks for N and C for max power	C cancels X , N matches R
is three-phase	per-phase method (Y equivalent), multiply power by 3
has a core with an air gap	magnetic circuit: $NI = \Phi(\mathcal{R}_c + \mathcal{R}_g)$
gives a B–H table	read B from the table, don't use a fixed μ
is a DC motor	draw I_L , I_f , I_a ; KVL with E_a ; speed ratios
is an induction motor	$n_s = 120f/p$, slip, sP_{ag} and $(1 - s)P_{ag}$

What to memorize (not on the formula sheet)

1/2

Power sign

$p = vi$ with i into + (absorbed)

Nodal / mesh

by-inspection matrices, supernode, supermesh

Norton

$I_N = I_{sc}$, $R_{Th} = V_{oc}/I_{sc}$

Transients

$x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$; $\tau = RC, L/R$

RLC

series $\alpha = R/2L$, parallel $\alpha = 1/2RC$, $\omega_0 = 1/\sqrt{LC}$

RMS

sinusoid $V_m/\sqrt{2}$; pulse $\sqrt{D}V_m$; sawtooth $V_m/\sqrt{3}$

Phasors

$\sin x = \cos(x - 90^\circ)$; $d/dt \rightarrow j\omega$

AC power

$P = V_{rms}I_{rms} \cos \theta$, $Q = \dots \sin \theta$, $\mathbf{S} = \mathbf{VI}^*$

PF correction

$Q_C = P(\tan \theta_1 - \tan \theta_2)$, $C = Q_C/(\omega V_{rms}^2)$

AC max power

$\mathbf{Z}_L = \mathbf{Z}_{Th}^*$, $P_{max} = V_{Th,rms}^2/4R_{Th}$

What to memorize (not on the formula sheet)

2/2

Transformer

$$V_2/V_1 = N_2/N_1, I_2/I_1 = N_1/N_2, \mathbf{Z}' = \mathbf{Z}_L(N_1/N_2)^2$$

Three-phase

$$\text{Y: } V_L = \sqrt{3}V_p; \Delta: I_L = \sqrt{3}I_p; P = \sqrt{3}V_L I_L \cos \theta; \mathbf{Z}_Y = \mathbf{Z}_\Delta/3$$

Magnetics

$$B = \mu H, \Phi = BA, \mathcal{R} = \ell/\mu A, NI = \Phi \mathcal{R}, L = N^2/\mathcal{R}, \mu_0 = 4\pi \times 10^{-7}$$

Induced voltage

$$e = N d\Phi/dt; E_{rms} = 4.44fN\Phi_{max}; e = B\ell u; f = B\ell v$$

DC machines

$$E_a = k\Phi\omega, T = k\Phi I_a, \text{ motor } V = E_a + I_a R_a$$

AC machines

$$n_s = 120f/p, s = (n_s - n)/n_s$$

Mechanics

$$P = T\omega, \omega = 2\pi n/60, 1 \text{ hp} = 746 \text{ W}$$

Exam-day checklist

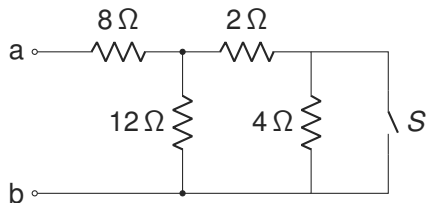
- Approved calculator, in **degree** mode for phasors, and practiced in complex mode.
- Read each question twice: method required? peak or RMS? line or phase? which turns ratio?
- Draw the circuit you're actually solving (redraws, per-phase circuit, reflected circuit, magnetic circuit).
- Write equations with symbols first, then numbers.
- Box every answer with units, direction, and lagging/leading.
- Use the free checks: KCL, KVL, power balance, sanity of sizes.
- Stuck? Write the method and the equations anyway. Method marks add up.

Final check

12 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

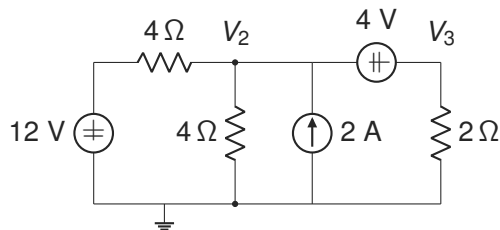
R1: equivalent resistance with a switch



Find R_{ab} with S open and with S closed. (S is across the $4\ \Omega$.)

your work here ↓

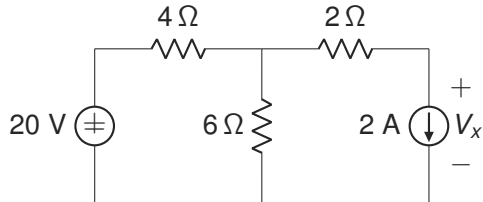
R2: supernode



The 4 V source has $+$ on the right. Find V_2 and V_3 (nodal).

your work here ↓

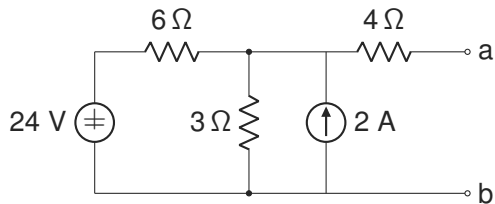
R3: mesh with a current source



Find I_1 (left mesh, clockwise)
and V_x (mesh).

your work here ↓

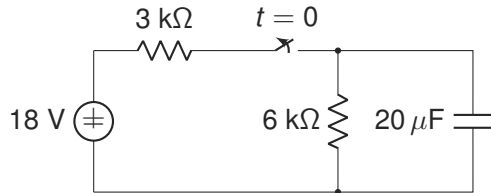
R4: Thévenin and max power



Thévenin at a–b, R_L for max power, P_{max} .

your work here ↓

R5: RC transient



Closed for a long time, opens at $t = 0$. Find $v_C(t)$ and $v_C(0.24 \text{ s})$.

your work here ↓

R6: series RLC in AC

$v_s = 100 \cos(500t)$ V drives $R = 40 \, \Omega$, $L = 0.12$ H and $C = 50 \, \mu\text{F}$ in series.

Find $\mathbf{Z}$, $i(t)$, the power factor and the average power.

your work here ↓

R7: power factor correction

A 30 kW load at pf 0.75 lagging is on 240 V RMS, 60 Hz. Find the parallel capacitor for pf 0.95 lagging.

your work here ↓

R8: transformer matching

A source (30 V RMS, $R_s = 450 \, \Omega$) must drive an $8 \, \Omega$ load through a transformer. Turns ratio for maximum power, and that power?

your work here ↓

R9: three-phase Δ load

A balanced Δ load, $\mathbf{Z}_{\Delta} = 18 + j24 \Omega$, on a 240 V (line) system. Find I_p , I_L , P , Q and pf.

your work here ↓

R10: magnetic circuit

Iron path 0.6 m, $\mu_r = 2500$, cross-section 5 cm², air gap 1.5 mm, $N = 400$.

Current for $B = 1.2$ T, and the inductance.

your work here ↓

R11: DC shunt motor

Rated 220 V, 1000 rpm, 15 hp, efficiency 85%, $R_f = 110\ \Omega$, $R_a = 0.2\ \Omega$.

Find I_L , I_a , E_a , and the speed when I_a falls to 25 A (constant flux).

your work here ↓

R12: induction motor

An 8-pole, 60 Hz induction motor runs with 2.5% slip and 20 kW crossing the air gap.

Find its speed, the rotor frequency, the rotor copper loss and the mechanical power.

your work here ↓

Answer sheet

Full worked solutions to P1–P12.

R1: solution

Read it S connects the node between the $2\ \Omega$ and $4\ \Omega$ to the bottom wire, so it sits across the $4\ \Omega$ **only**

Open a gap: right branch $2 + 4 = 6\ \Omega$; it's between the same nodes as the $12\ \Omega$: $12 \parallel 6 = 4$; plus the input $8\ \Omega$: $12\ \Omega$

Closed a wire across the $4\ \Omega$ shorts it out ($0\ \text{V}$, no current): right branch is just $2\ \Omega$

$$12 \parallel 2 = \frac{24}{14} = 1.71\ \Omega, \text{ plus } 8: 9.71\ \Omega$$

Trap: a switch across the *whole* right branch would short both the 2 and the 4, giving $8\ \Omega$. Always check exactly which two nodes it joins.

R2: solution

Plan $V_1 = 12\text{ V}$ is fixed (source to ground). The 4 V source floats between V_2 and V_3 : supernode

Source eq. + on the V_3 side: $V_3 = V_2 + 4$

Supernode all currents leaving the bubble:
$$\frac{V_2 - 12}{4} + \frac{V_2}{4} - 2 + \frac{V_3}{2} = 0 \quad (\text{the } 2\text{ A enters: } -2)$$

Substitute $\times 4$: $V_2 - 12 + V_2 - 8 + 2(V_2 + 4) = 0 \Rightarrow 4V_2 = 12$

Result

$$V_2 = 3\text{ V}$$

$$V_3 = 7\text{ V}$$

Check $-2.25 + 0.75 - 2 + 3.5 = 0 \checkmark$

R3: solution

Known mesh the 2 A source is on mesh 2's outer edge pointing down (clockwise): $I_2 = 2 \text{ A}$

Mesh 1 walk clockwise: up the 20 V (+20), across the 4Ω ($-4I_1$), down the shared 6Ω ($-6(I_1 - I_2)$)

Solve $20 - 4I_1 - 6I_1 + 12 = 0 \Rightarrow 10I_1 = 32 \Rightarrow I_1 = 3.2 \text{ A}$

V_x mesh 2 clockwise including the source's unknown voltage: $-2(2) - V_x - 6(2 - 3.2) = 0$

Result $-4 - V_x + 7.2 = 0 \Rightarrow V_x = 3.2 \text{ V}$ (top +)

R4: solution

R_{Th} kill the sources ($24\text{ V} \rightarrow \text{wire}$, $2\text{ A} \rightarrow \text{gap}$): $6 \parallel 3 = 2$, plus the series 4: $R_{Th} = 6\ \Omega$

V_{oc} load removed: no current in the $4\ \Omega$, so V_{Th} is the middle node voltage V

KCL
$$\frac{V - 24}{6} + \frac{V}{3} = 2 \xrightarrow{\times 6} V - 24 + 2V = 12 \Rightarrow$$

$$V_{Th} = 12\text{ V}$$

Max power load matched to R_{Th} : $R_L = 6\ \Omega$,

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{144}{24} = 6\text{ W}$$

R5: solution

$t = 0^-$ closed a long time: C $\rightarrow$ gap; divider: $v_C = 18 \cdot \frac{6}{3+6} = 12 \text{ V}$

0^+ continuity: $v_C(0^+) = 12 \text{ V}$

After switch open: source and 3 k disconnected, C discharges through the 6 k: $v_C(\infty) = 0$

τ resistance seen by the C: 6 k: $\tau = (6000)(20 \times 10^{-6}) = 0.12 \text{ s}$

$v_C(t)$ $0 + (12 - 0)e^{-t/\tau}$: $12 e^{-t/0.12} \text{ V}$

0.24 s $t = 2\tau$: $12e^{-2} = 12 \times 0.135 = 1.62 \text{ V}$

R6: solution

Impedances $\omega = 500$: $\mathbf{Z}_L = j(500)(0.12) = j60$,

$$\mathbf{Z}_C = \frac{-j}{(500)(50 \times 10^{-6})} = -j40 \Omega$$

Z series: $40 + j60 - j40 = 40 + j20 \Omega = 44.7 \angle 26.6^\circ$

I $\frac{100 \angle 0^\circ}{44.7 \angle 26.6^\circ} = 2.24 \angle -26.6^\circ \text{ A} \Rightarrow$

$$i(t) = 2.24 \cos(500t - 26.6^\circ) \text{ A}$$

pf $\cos 26.6^\circ = R/|\mathbf{Z}| = 40/44.7 = 0.894 \text{ lagging } (X > 0)$

P source given as a **peak** value, so

$$P = \frac{1}{2} |\mathbf{I}|^2 R = \frac{1}{2} (2.24^2)(40) = 100 \text{ W}$$

R7: solution

Before $\theta_1 = \cos^{-1} 0.75 = 41.4^\circ$: $Q_1 = 30 \tan 41.4^\circ = 26.5 \text{ kvar}$

Target $\theta_2 = \cos^{-1} 0.95 = 18.2^\circ$: $Q_2 = 30 \tan 18.2^\circ = 9.86 \text{ kvar}$

Capacitor supplies the difference: $Q_C = 26.5 - 9.86 = 16.6 \text{ kvar}$

$$C \quad \frac{Q_C}{\omega V_{rms}^2} = \frac{16\,600}{(377)(240^2)} = 764 \mu\text{F}$$

R8: solution

Goal the $8\ \Omega$ load, reflected to the primary, must look like
 $R_s = 450\ \Omega$

Ratio $(N_1/N_2)^2 \times 8 = 450 \Rightarrow N_1/N_2 = \sqrt{56.25} = 7.5 : 1$

Power matched source: $P_{max} = \frac{V_s^2}{4R_s} = \frac{30^2}{4 \times 450} = 0.5\ \text{W}$ (all delivered to the load)

R9: solution

I_p Δ : each impedance sees the line voltage:

$$240/|18 + j24| = 240/30 = 8 \text{ A}$$

$$I_L \text{ } \Delta \text{ line current: } \sqrt{3} \times 8 = 13.9 \text{ A}$$

$$P \quad 3I_p^2 R = 3 \times 64 \times 18 = 3456 \text{ W}$$

$$Q \quad 3I_p^2 X = 3 \times 64 \times 24 = 4608 \text{ var}$$

$$\text{pf} \quad R/|Z| = 18/30 = 0.6 \text{ lagging}$$

R10: solution

$$\Phi \quad BA = 1.2 \times 5 \times 10^{-4} = 0.6 \text{ mWb}$$

$$\mathcal{R}_{core} \quad \frac{0.6}{2500(4\pi \times 10^{-7})(5 \times 10^{-4})} = 3.82 \times 10^5 \text{ A-t/Wb}$$

$$\mathcal{R}_{gap} \quad \frac{1.5 \times 10^{-3}}{(4\pi \times 10^{-7})(5 \times 10^{-4})} = 2.39 \times 10^6 \text{ A-t/Wb (gap dominates)}$$

I series total 2.77×10^6 :

$$I = \frac{\Phi \mathcal{R}}{N} = \frac{0.6 \times 10^{-3} \times 2.77 \times 10^6}{400} = 4.15 \text{ A}$$

$$L \quad N^2/\mathcal{R} = 400^2/(2.77 \times 10^6) = 57.8 \text{ mH}$$

R11: solution

P_{in} rated 15 hp is output: $\frac{15 \times 746}{0.85} = 13\,165\text{ W},$

$$I_L = 13\,165/220 = \boxed{59.8\text{ A}}$$

I_a field: $I_f = 220/110 = 2\text{ A},$ so $I_a = 59.8 - 2 = \boxed{57.8\text{ A}}$

E_a armature KVL: $220 - 57.8(0.2) = \boxed{208.4\text{ V}}$

New E_a $I_a = 25\text{ A}: 220 - 25(0.2) = 215\text{ V}$

Speed constant flux, speed $\propto E_a$: $1000 \times \frac{215}{208.4} = \boxed{1032\text{ rpm}}$

R12: solution

$$n_s \quad 120f/p = 120(60)/8 = 900 \text{ rpm}$$

$$\text{Speed} \quad (1 - s)n_s = 0.975 \times 900 = 877.5 \text{ rpm}$$

$$\text{Rotor } f \quad sf = 0.025 \times 60 = 1.5 \text{ Hz}$$

$$\text{Rotor loss} \quad sP_{ag} = 0.025 \times 20 = 0.5 \text{ kW}$$

$$\text{Mechanical} \quad (1 - s)P_{ag} = 0.975 \times 20 = 19.5 \text{ kW}$$